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NATURAL LANGUAGE PROCESSING
PREDICTING **REAL** VS FAKE NEWS

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Project Objective & Dataset

From Headlines to Classification

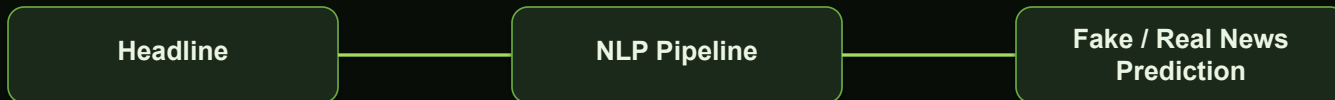
We built a binary text classifier to distinguish real news from fake news using only the headline text.

File	Purpose
training_data.csv	Labelled data used for training and validation
testing_data.csv	Unlabelled final dataset with placeholder label 2
testing_data_predictions.csv	Final output with predicted labels 0 or 1

Label	Meaning
0	Fake news
1	Real news

Key Constraint:

We are classifying short headlines, not full articles. This matters because short text often depends heavily on keywords, phrasing, and limited context.



Classical NLP Pipeline

Classical NLP Workflow

Raw text:

- text cleaning
- stopword removal
- lemmatisation
- train/test split
- vectorisation (normalisation for BOW only)
- classifier
- evaluation

Preprocessing step	Purpose
Text cleaning	Remove noise, punctuation, symbols, duplicates, inconsistent formatting
Stopword removal	Remove frequent low-information words
Lemmatisation	Morphological analysis of words
Train/Test split	Evaluate on unseen labelled data
Vectorisation	Convert text into numerical features
Normalisation	Normalise the vectors

Machine learning models cannot read text directly. We first had to convert headlines into numbers.



Preprocessing Insight

Preprocessing Is Not Neutral

During cleaning, we found that a few very short headlines became empty after stopwords removal.

Very short headline cases:

- mostly stopwords
- stopwords removed
- empty string

	label	text	text_clean	text_no_stopwords	text_lemm
986	0	no	no		
1708	0	once again	once again		
8891	0	so	so		

Column	Meaning
text	Original headline
text_clean	Cleaned headline
text_no_stopwords	Stopwords removed
text_lemmatized	Final Classical NLP input

Why this matters:

- Text preprocessing can improve signal.
- But it can also remove too much information.
- Keeping intermediate columns helped us understand where information was lost.

Decision: Rows with empty processed text were removed before modelling.

This was a useful quality-control moment: we did not just run models; we inspected how preprocessing changed the data.

Vectorisation Experiments

CountVectorizer vs TF-IDF across classical model baselines

Turning Headlines into Features

We compared raw word-count features against TF-IDF weighting to understand which representation worked best for short headline classification.

Model / Experiment	CountVectorizer accuracy	TF-IDF accuracy	Better Vectoriser	Observation
Random Forest baseline	91.52%	91.12%	CountVectorizer	CountVectorizer performed slightly better with default unigrams.
MultinomialNB baseline	92.77%	92.66%	CountVectorizer	Very close, but CountVectorizer had the edge.
XGBoost baseline	88.89%	87.94%	CountVectorizer	CountVectorizer performed better by almost 1 percentage point.
Random Forest bigram setup	90.79%	90.84%	TF-IDF	TF-IDF was marginally better here, but both were below the unigram baseline.
Random Forest custom tokenizer	90.73%	90.73%	Tie	No meaningful difference.

Main Finding

CountVectorizer generally performed better than TF-IDF across the classical models, especially with unigram/default setups.

Interpretation

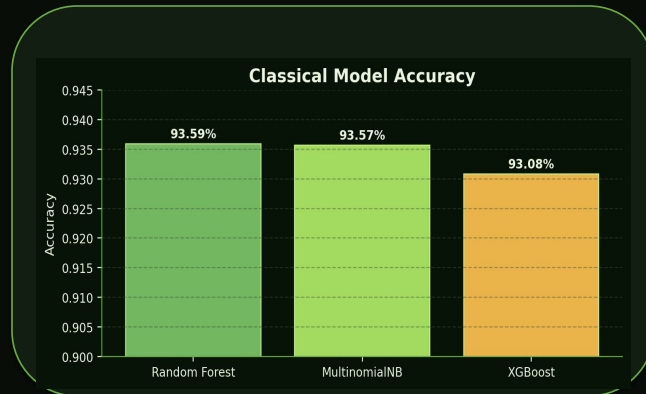
Because headlines are short, simple word and phrase presence was often more useful than weighting words by document rarity.

Classical Models Compared

Benchmarking Random Forest, MultinomialNB and XGBoost

Classical Model Benchmark

Model	Best Feature Setup	Accuracy	Key Observation
Random Forest	Tuned CountVectorizer setup	93.59%	Marginally highest uploaded classical score, but more complex and costlier.
MultinomialNB	CountVectorizer, unigrams + bigrams	93.57%	Near-identical performance; fast, simple and interpretable.
XGBoost	Tuned CountVectorizer	93.08%	Strong after wider tuning, but below NB/RF.



Random Forest

Marginally highest classical score in the uploaded notebooks.

Good performance, but more computationally expensive.

Less transparent than MultinomialNB for explaining text-count patterns.

MultinomialNB

Near-identical accuracy to Random Forest: 93.57%.

Low complexity, fast training and easy interpretation.

Strong fit for CountVectorizer text-count features.

XGBoost

Improved after wider tuning, reaching 93.08%.

Required more hyperparameter exploration.

Showed that model complexity alone did not guarantee the best result.

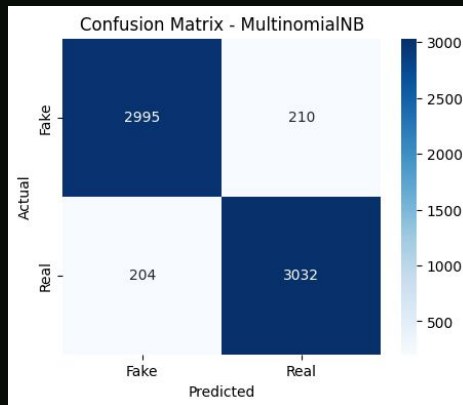
Takeaway: Random Forest and MultinomialNB were effectively tied at the top. MultinomialNB remained the most practical classical reference because it delivered almost the same accuracy with simpler, faster and more explainable modelling. Feature representation and careful tuning mattered more than model complexity alone.

MultinomialNB: Balanced Classical Reference

MultinomialNB Fine-Tuning

Component	Best setting
Vectoriser	CountVectorizer
ngram_range	(1, 2)
max_df	1.0
min_df	1
max_features	None
alpha	0.5
fit_prior	True

Metric	Result
Accuracy	93.57%
Macro F1	93.57%
Weighted F1	93.57%



	Pred Fake	Pred Real
Actual Fake	2995	210
Actual Real	204	3032

Interpretation:

The model was balanced across both classes.
False fake/real errors were similar.
It handled both fake and real headlines with similar reliability.

XGBoost and Random Forest: What We Learned

Tree-Based Models: Strong but Costlier

Random Forest:

Best baseline vectoriser: Bag of Words unigram setup.
Hyperparameter tuning used RandomizedSearchCV.
Tuned RF reached around 93.59% accuracy.
It performed very strongly but required more computation than MultinomialNB.

Best RF Setup

Parameter	Value
n_estimators	500
max_depth	None
min_samples_split	2
min_samples_leaf	1
max_features	log2
criterion	gini

XGBoost:

Started around 89.57% with default CountVectorizer setup.
Unigrams + bigrams worsened performance.
Only bigrams performed poorly.
Reducing features also reduced performance.
Wider tuning improved performance to 93.08%.

Best XGBoost Setup

Parameter	Value
gamma	0.5
learning_rate	0.2
max_depth	12
n_estimators	800

Main Lesson:

More complex models did not automatically win. Feature representation and careful tuning mattered more than model complexity alone.

Final Testing Data Predictions

From Validation to Final Output

After selecting the final classical model, we applied the same preprocessing pipeline to `testing_data.csv`.

`testing_data.csv`

- same cleaning
- same stopword removal
- same lemmatisation
- same fitted vectoriser
- final model prediction
- replace label 2 with 0/1
- save final CSV

Why this matters:

The testing data must go through the same transformation logic as the training data. Otherwise, the model receives data in a different format from what it learned.

Predicted label	Count	Share
0 - Fake	4,320	46.9%
1 - Real	4,889	53.1%

Transformer Extension

Moving Beyond Word Counts: Transformers

After building strong classical baselines, we tested Transformer-based models.

Classical NLP uses **manually created features**:

Word Counts / N-Grams / TF-IDF

Transformers learn **contextual representations**:

headline meaning → token embeddings → classification

Raw headline
- pipeline inference (DistilBERT)
- model tokenizer
- pretrained Transformer
- fine-tuning with Trainer
- validation evaluation

Important Distinction:

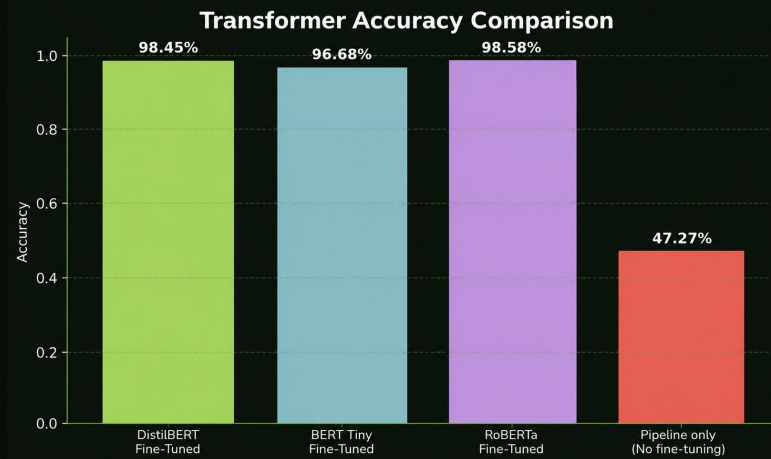
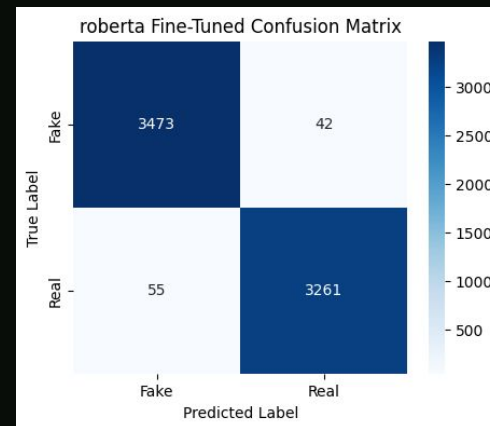
For Transformers, we **used the raw headline text, not the heavily cleaned text**. The tokenizer is designed to handle punctuation, casing, and subword structure.

Models Tested / Planned	Approach
DistilBERT base uncased	Pipeline + Fine-tuned with Hugging Face Trainer
BERT Tiny fake-news model	Fine-tuned with Hugging Face Trainer
RoBERTa fake news classification	Fine-tuned with Hugging Face Trainer

Transformer Results

Fine-Tuning Made the Difference

Transformer model	Approach	Accuracy	Key Point
DistilBERT base uncased	Pipeline	47.27%	Not adapted to task
	Fine-tuned with Trainer	98.45%	Excellent results
BERT Tiny fake-news model	Fine-tuned with Trainer	96.68%	Smaller and faster, still strong
RoBERTa fake news classification	Fine-tuned with Trainer	98.58%	Best uploaded Transformer result



Main Insight:
The same architecture family can perform poorly without task adaptation, but very strongly after fine-tuning. Highly dependent on model training data.

Fine-tuning turned a general language model into a fake-news headline classifier.

Classical NLP vs Transformers

What Changed When We Used Transformers?

Aspect	Classical NLP	Transformers
Input	Cleaned and lemmatised text	Raw headlines
Feature creation	Manual vectorisation	Learned representations
Main tools	CountVectorizer, TF-IDF	Tokenizer, pretrained model
Best performance	~93.6%	Up to ~98.6%
Cost	Lower	Higher compute
Interpretability	Higher	Lower
Best use	Fast, strong baseline	Higher performance with fine-tuning

Core learning: Classical NLP gave us a reliable, interpretable baseline. Transformers improved performance by learning richer contextual patterns, but required more compute and more careful setup.

Final Conclusions

What We Learned

1. Preprocessing matters

Cleaning improved consistency, but also showed that aggressive preprocessing can remove short headlines completely.

2. Simple models can be powerful

MultinomialNB with CountVectorizer reached 93.57%, proving that strong baselines matter.

3. Complexity is not automatically better

XGBoost required significant tuning and still did not beat the strongest classical baselines.

4. Fine-tuning is the Transformer turning point

A pipeline-only Transformer performed poorly, while fine-tuned Transformers reached 96-98%+.

5. The final workflow was end-to-end

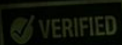
We moved from raw data to validated models to final prediction files.

We started with strings, broke them into features, then used Transformers to learn deeper context - hence, **Breaking Bad Strings**.

Daily Chronicle

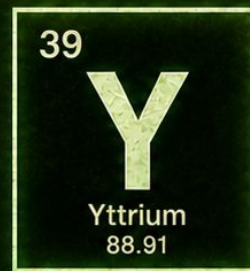
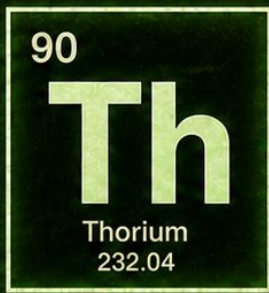
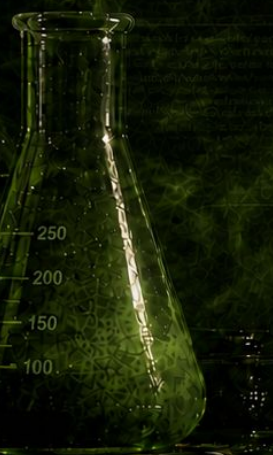
TRUTH. VERIFIED. FACTS MATTER.

EST. 1921



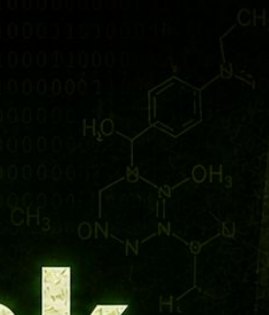
FACTS BACKED BY EVIDENCE

Science confirms climate patterns.



Thank You

CHEMISTRY OF TRUTH. BREAKING BAD STRINGS.



DAILY RUMOR
TODAY: 10/27

BREAKING!!!

SHOCKING TRUTH THEY DON'T WANT YOU TO KNOW!

SECRET EXPOSED
YOU WON'T BELIEVE WHAT HAPPENED

- UNBELIEVABLE
- NEW CLAIM
- VIRAL & TRUE?

UNCHECKED X

FACT OR FICTION?
SOCIAL MEDIA BUZZ EXPLODES
DOF. JUST SHARES.

CLASSIFIED
+ EYES ONLY